CAP 6545 · PROJECT PROPOSAL

Do Conformal Prediction Intervals Stay Valid Under Distribution Shift?

A leak-free study of protein–ligand binding affinity prediction

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THE PROBLEM

01	Drug discovery needs to know how tightly a molecule binds a protein (binding affinity)
02	Deep learning models predict it, but give one number with no confidence
03	Conformal prediction wraps any model in an interval with a guarantee: e.g. 90% of true values fall inside
04	**The guarantee assumes calibration and test data look alike. In drug discovery, new targets and new chemistry break that assumption**

WHAT IS KNOWN, AND THE GAP

KNOWN

Conformal prediction has been applied to binding affinity: valid on one external target, not on another (Parks et al., 2020)

Weighted conformal prediction corrects covariate shift (Tibshirani et al., NeurIPS 2019); adapted to drug properties (CoDrug, NeurIPS 2023)

Uncertainty for structure-based models degrades under shift on leak-free splits, but without conformal methods (Rayka & Naghavi, Sci. Rep. 2025)

GAP

No conformal evaluation I found on leak-free LP-PDBBind, BDB2020+, and a peptide-ligand shift set.

This is an empirical extension, not a new method.

PILOT (WEAK PROXY MODEL, REAL LP-PDBBIND DATA): TEST COVERAGE AT A 90% TARGET

90.3%
calibrated on the LP validation set

82.3%
calibrated on a random slice of training data

89.8%
after normalizing by model uncertainty

RESEARCH QUESTIONS

Q1 How does the calibration set (random training hold-out vs. LP-PDBBind validation) change coverage on the LP test set, BDB2020+, and the peptide set?

Q2 Does normalizing intervals by each method's uncertainty keep coverage at target under shift, while staying narrow?

Q3 Does weighted conformal prediction recover the coverage lost under shift?

MODEL Feed-forward network on ECIF features (Sánchez-Cruz et al., Bioinformatics 2021), with deterministic, deep ensemble, MC dropout, and Bayes-by-Backprop versions

DATA AND RESOURCES

SET	ROLE	SIZE
LP-PDBBind (Li et al., J. Phys. Chem. B 2026)	train / validation / test	7,259 / 947 / 2,166
BDB2020+	shifted test	111
Peptides-Holdout (Durant et al., Bioinformatics 2025)	shifted test	2,498

All public. I removed the 134 training complexes that overlap the peptide set, which reproduces the published training count of 7,259 exactly

PDBbind structure files: free for academics after registration

Compute: HiPerGator GPUs for ensembles and repeated seeds

Limit: BDB2020+ is small, so coverage there is only detectable within about ±5.6 points

PLAN AND RISKS

WEEKS	PLAN
Sep 29 – Oct 12	Register for PDBbind, build splits, compute ECIF features
Oct 13 – Oct 26	Train the baseline network, add ensemble, MC dropout, Bayes-by-Backprop
Oct 27 – Nov 9	Split conformal intervals and uncertainty-normalized scores (Q1, Q2)
Nov 10 – Nov 23	Weighted conformal (Q3), selective-prediction analysis, start report
Nov 24 – Dec 8	Final presentation (Dec 1 week), report and code due Dec 8, 3 PM

RISKS AND FALLBACKS

Time: drop Bayes-by-Backprop first, then the selective-prediction analysis

Weighted conformal may be unstable: report that as a finding

Possible upgrade if results are interesting: conformal selection with false-discovery-rate control

REFERENCES

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